

CHE 396 Quiz 1

Allison Nordvall

TOTAL POINTS

86.5 / 100

QUESTION 1

11 15 / 15

- 3 pts no detail explanation about the process
- 1 pts no applications mentioned
- ✓ - 0 pts correct
- 15 pts incorrect

QUESTION 2

22 15 / 15

- 12 pts other processes not mentioned
- 3 pts no brief information about the processes
- ✓ - 0 pts correct

QUESTION 3

33 15 / 15

- 10 pts Major logical error in the BFD
- 5 pts Minor mistake in the BFD
- ✓ - 0 pts correct

QUESTION 4

44 15 / 15

- 15 pts separation process not mentioned
- 10 pts Partial process
- 5 pts Minor error
- ✓ - 0 pts correct

QUESTION 5

55 5 / 5

- 2.5 pts partial reaction
- ✓ - 0 pts correct
- 5 pts incorrect

QUESTION 6

66 5 / 5

- ✓ - 0 pts Correct
- 5 pts wrong technique

- 2.5 pts partially correct

QUESTION 7

77 1.5 / 5

- 5 pts No full form or production process
- ✓ - 3.5 pts full form but no production process
- 0 pts correct
- 2 pts partially production process
- 1.5 pts no full form

QUESTION 8

88 5 / 5

- ✓ - 0 pts Correct
- 2 pts Partially correct
- 4 pts wrong
- 5 pts not written

QUESTION 9

99 10 / 20

- 10 pts incorrect logic for recovery rate (step 1)
- 10 pts incorrect logic for recovery rate (step 2)
- 0 pts correct
- ✓ - 5 pts correct logic but calculation error for recovery rate (step 1)
- ✓ - 5 pts correct logic but calculation error for recovery rate (step 2)

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1. What is the design basis for your mini-project?

The design basis was to add an absorber and a Vent to decrease the amount of acetone lost when removing hydrogen gas. This would be done by increasing the pressure from 2.6 to 17 atm in the absorber that was added.

2. List competing processes.

1. Turton (we modified this by adding an absorber and a vent)

- Dehydrogenization of IPA

2. ARJE process

(Acetone, butanol, and ethanol)

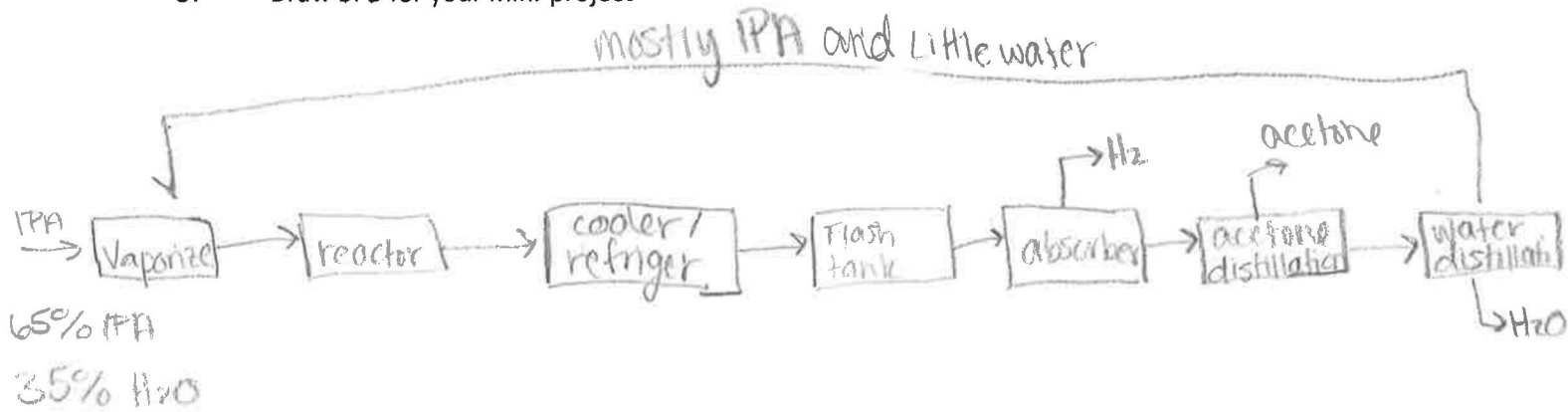
- This is more of a separation process using 3 different distillation columns

3. Cumene process

- Most commonly used

- acetone is a coproduct of propanol.

3. Draw BFD for your mini-project



4. Explain separation section of your mini-project in details.

There are technically 3 parts even though there are only two distillation columns.

- Hydrogen gas is separated out in the absorber. Slight amounts of acetone are removed in this section
- Acetone of 99.9% purity is then separated using a distillation column. The rest of the IPA and water will be fed to the second distillation column.
- The second distillation column separates water and IPA. The IPA that is separated is then recycled to the beginning of the process where it can be used again.

5. What is monoclonal antibody and its main usage?

A monoclonal antibody is a clone of a cell that fights against certain diseases or cancers. The good cells that can fight want to be cloned to rid the body of certain cancers or diseases.

6. Explain how hydrogen is separated by PSA? → Pressure swing

1st air is separated at high pressure using a solid.

This solid will absorb the hydrogen and let the Nitrogen pass through. The inflow can then be stopped. The pressure is then swung down so that the hydrogen can be released from the solid and be collected.

7. What is formalin? Explain briefly its production.

Formalin is a very diluted version of formaldehyde. This is the form that formaldehyde is used in. Formalin is well known to be used in shampoos and many other things.

8. How ethanol is produced in chemical industry?

Corn is mostly used to produce ethanol in the chemical industry. It is heated then pressurized then distilled to remove all impurities.

Ethylene - 28 g/mol

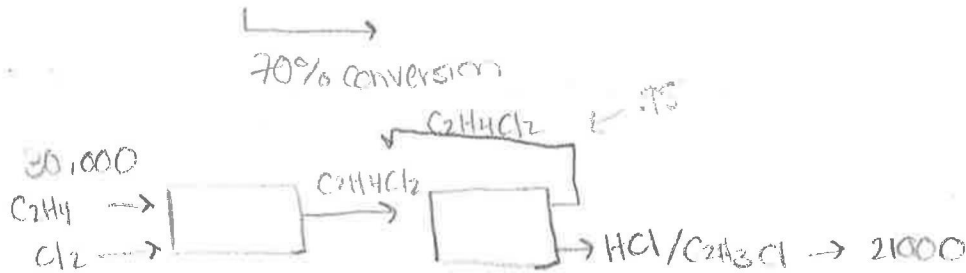
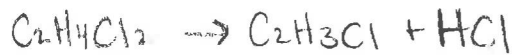
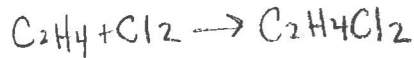
Cl₂ - 71 g/mol

HCl - 36.5 g/mol

Vinyl chloride - 62.5 g/mol

Dichloroethane - 99 g/mol

9. 30,000 lb/hr of Ethylene (C₂H₄) enters direct chlorination unit with sufficient amount of Cl₂ gas and produces dichloroethane (C₂H₄Cl₂) with %100 conversion. Then, dichloroethane reacts in pyrolysis unit with a conversion of 70% producing vinyl chloride and HCl. After pyrolysis, dichloroethane is separated by distillation with a recovery rate of 95% and unreacted dichloroethane is recycled back to the pyrolysis. Calculate the vinyl chloride output flow rate and recycle stream flow rate in lb/hr.



Recycle stream can be calculated using the equation

$$30,000 \text{ lb/hr} \times (.7) = 21,000 \text{ lb/hr vinyl chloride / HCl}$$

$$(30,000 - R)(.95) = 21,000$$

$$R = 7894.737 \text{ lb/hr}$$

Recycle stream flow rate

Vinyl Chloride / HCl had an outflow of 21,000 lb/hr